

Submission PHYS1-MID-SUB01

Question PHYS1-MID-Q01

scratch: show step

1. $v = v_i + at = 4 + 3(5) = 19 \text{ m/s}$

2. $\Delta x = v_i t + \frac{1}{2}at^2 = 4(5) + \frac{1}{2}(3)(5)^2 = 57.5 \text{ m}$

=> FINAL: $v = 19 \text{ m/s}$; displacement = 57.5 m (check units/interpretation)

Question PHYS1-MID-Q02

scratch: show step

1. $N = mg = 58.8 \text{ N}$; friction = $\mu N = 11.8 \text{ N}$

2. $\Sigma F = 17 - 11.8 = 5.2 \text{ N}$

3. $a = \Sigma F / m = 0.87 \text{ m/s}^2$

=> FINAL: $a = 0.87 \text{ m/s}^2$; friction opposes motion (check units/interpretation)

Question PHYS1-MID-Q03

scratch: show step

1. $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2 \cdot 9.8 \cdot 8}$

2. $v = 12.52 \text{ m/s}$; mass cancels

=> FINAL: $v = 12.52 \text{ m/s}$ (check units/interpretation)

Question PHYS1-MID-Q04

scratch: show step

1. $p_{\text{total, before}} = p_{\text{total, after}}$

2. It applies when external impulse on the system is negligible.

=> FINAL: total momentum before equals after when external impulse is negligible (check units/interpretation)

Question PHYS1-MID-Q05

scratch: show step

1. Impulse is area under $F(t)$: $J = F\Delta t = 6(0.5) = 3 \text{ N}\cdot\text{s}$.

2. $\Delta p = J$, so momentum changes by $3 \text{ kg}\cdot\text{m/s}$ in the force direction.

=> FINAL: impulse = $3 \text{ N}\cdot\text{s}$; momentum changes by $3 \text{ kg}\cdot\text{m/s}$ (check units/interpretation)

Question PHYS1-MID-Q06

scratch: show step

1. $v = v_i + at = 7 + 3(2) = 13 \text{ m/s}$

2. $\Delta x = v_i t + \frac{1}{2}at^2 = 7(2) + \frac{1}{2}(3)(2)^2 = 20 \text{ m}$

=> FINAL: $v = 13 \text{ m/s}$; displacement = 20 m (check units/interpretation)

Question PHYS1-MID-Q07

scratch: show step

1. $N = mg = 19.6 \text{ N}$; friction = $\mu N = 3.9 \text{ N}$

2. $\Sigma F = 16 - 3.9 = 12.1 \text{ N}$

3. $a = \Sigma F / m = 6.04 \text{ m/s}^2$

=> FINAL: $a=6.04 \text{ m/s}^2$; friction opposes motion (check units/interpretation)

Question PHYS1-MID-Q08

scratch: show step

1. $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2 \cdot 9.8 \cdot 9}$

2. $v = 13.28 \text{ m/s}$; mass cancels

=> FINAL: $v = 13.28 \text{ m/s}$ (check units/interpretation)